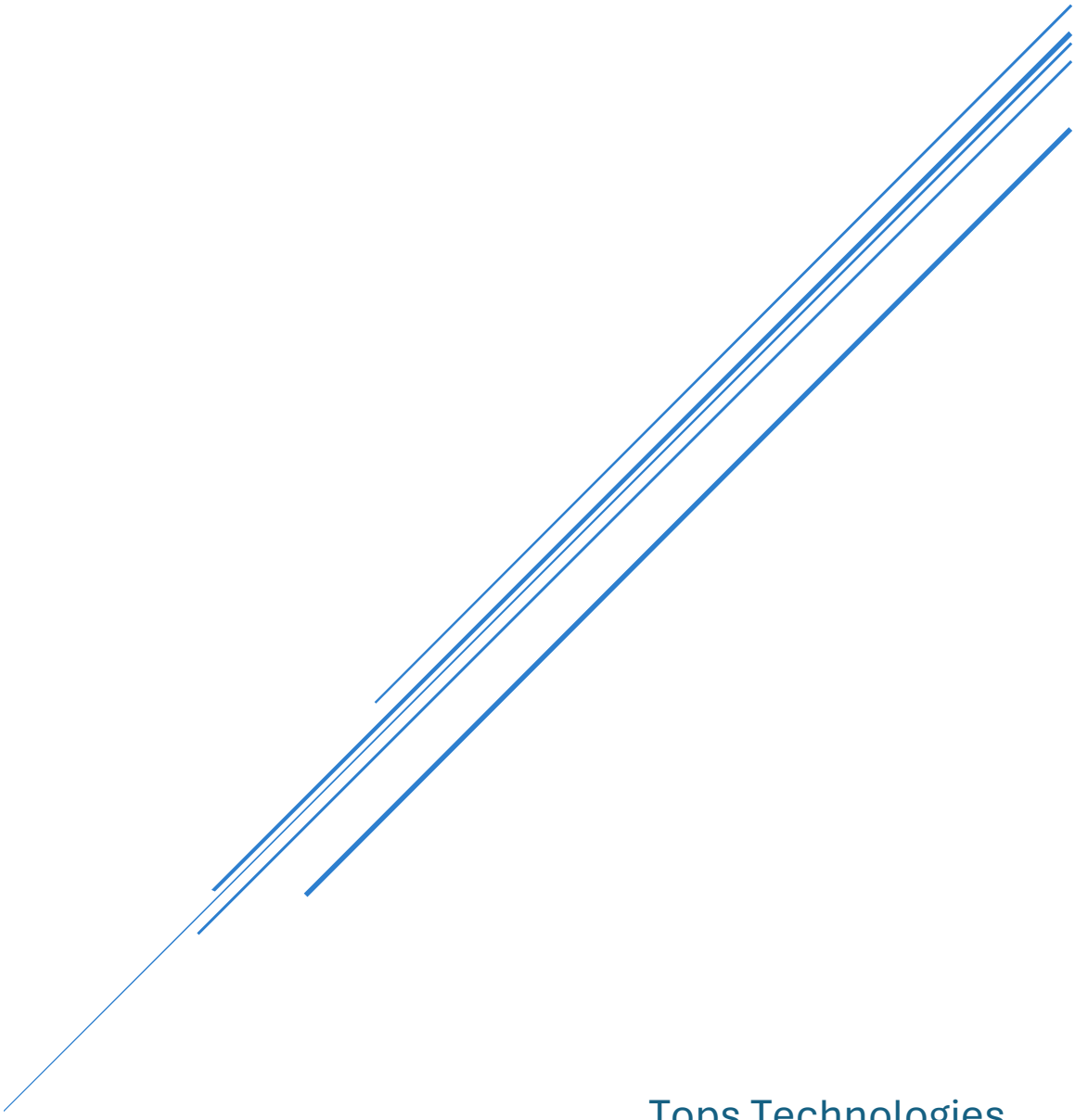
NETWORK DEVICES

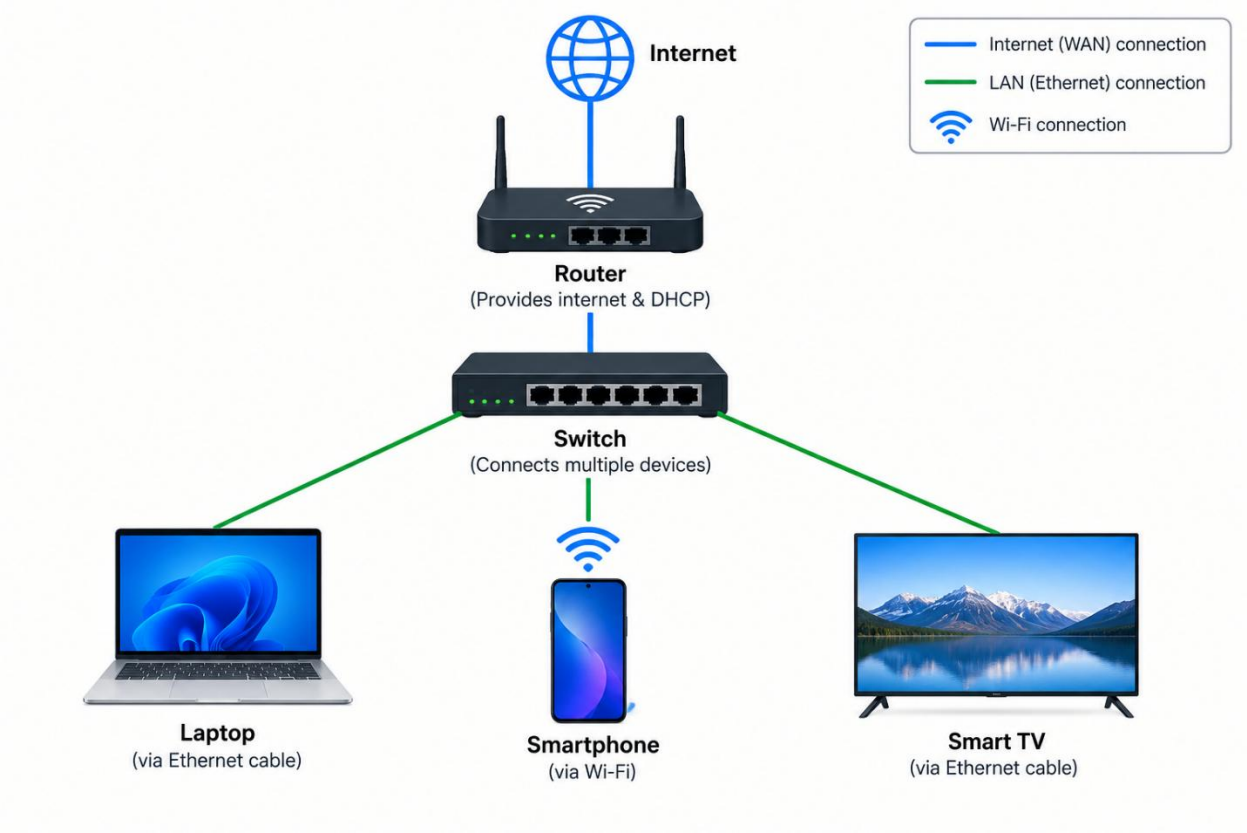
Assignment - 6



Tops Technologies
Cyber Security

1. **Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.**

➢



Network connections:

- **Router: Connects the home network to the Internet and provides IP addresses using DHCP.**
- **Switch: Connects multiple wired devices to the network.**
- **Laptop: Connected to the switch using an Ethernet cable.**
- **Smartphone: Connected wirelessly to the router through Wi-Fi.**
- **Smart TV: Connected to the switch using an Ethernet cable.**

2. **Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).**

➢

Network Device	Main Function	Real-World Use Case
Router	Connects different networks and provides Internet access.	I use a Wi-Fi router to connect my smartphone to the Internet and watch videos on YouTube.
Switch	Connects multiple devices within a LAN (Local Area Network).	In a computer lab, a switch connects multiple computers so students can play LAN games or share files.
Hub	Connects multiple devices and broadcasts incoming data to all connected ports.	In an old computer lab, a hub can connect several computers for basic local network communication.
Access Point (AP)	Provides wireless Wi-Fi connectivity to devices.	In a college, an access point allows students to connect their smartphones and laptops to Wi-Fi to use Google Classroom.
Modem	Connects a home or office network to an Internet service provider.	A modem helps provide an Internet connection so I can browse websites and use WhatsApp.

3. **Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).**

➢

Scenario: Ordering Food on Zomato

When I open the Zomato app on my smartphone and order food, my phone sends a request through Wi-Fi to the access point, which provides wireless connectivity. The data then travels through the router, which forwards it toward the Internet. A switch helps transfer data between devices within the local network, such as in my home or the service provider's network. The request reaches Zomato's servers through the Internet, and the server sends back restaurant details, menu items, and order confirmation to my phone through the same network path.

4. **Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.**
   Hint: Think about devices like switches, powerline adapters, or WiFi extenders.

➢

There are two devices I can use to connect my gaming PC to the Internet:

1. Powerline Adapter

- **It uses the existing electrical wiring in my home to carry network data.**
- **I connect one adapter to the router using an Ethernet cable and plug it into a power socket.**
- **I plug the second adapter near my gaming PC and connect it to the PC using an Ethernet cable.**
- **Use: It provides a wired network connection without running a long Ethernet cable through the house.**

2. Wi-Fi Extender with an Ethernet Port

- **It extends the Wi-Fi signal from my router to an area farther away.**
- **I place the extender near my gaming PC and connect the PC to its Ethernet port using a cable.**
- **The extender receives the router's Wi-Fi signal and provides Internet access to the PC through Ethernet.**
- **Use: It allows my gaming PC to connect without needing a built-in Wi-Fi adapter.**